

PAMPANA SIVA SURYA RAM GOPAL

Software Engineer | AI & Automation | Full-Stack Development

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PROFESSIONAL SUMMARY

Python Full-Stack Developer skilled in **Django**, **React**, and **Azure OpenAI** with a focus on building scalable, AI-driven web applications. Experienced in backend APIs, frontend design, and database management. Completed automation testing training to enhance software quality and performance awareness. CCNA-certified with strong cloud and networking knowledge.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, HTML5, CSS3

Frameworks & Tools: Django, React, Selenium, Pytest, TestNG, JUnit

Cloud & AI: Azure OpenAI, AWS, GPT-4, Azure AI Search, LangChain, LangGraph, RAG

Databases: MySQL, Azure Cosmos DB

Version Control & IDEs: Git, VS Code, PyCharm, GitHub Copilot, Postman

Networking: CCNA Certified – Routing, Switching, TCP/IP, Network Troubleshooting

PROFESSIONAL EXPERIENCE

Software Test Engineer Trainee

Cisco Systems

Bengaluru, India

Nov 2024 – Present

- Automated web testing workflows using **Python**, **Selenium**, and **Pytest**, improving regression coverage by 40%.
- Developed a full-stack AI chatbot leveraging **Azure OpenAI**, **Django**, and **MySQL** for real-time query processing.
- Engineered intelligent schema retrieval using **Azure AI Search**, reducing token usage by 30%.
- Applied **CCNA** certified skills in routing, switching, TCP/IP, and network troubleshooting for system optimization.

KEY PROJECTS

AI-Powered Chatbot for Infrastructure Management

Cisco, 2024 – Currently Implementing

- Architected a multi-agent system with **Azure OpenAI: GPT-4.0** primary agent and **GPT-3.5-Turbo** SQL agent.
- Intelligent query routing across **RAG**, **SQL**, and **web-search** pipelines using **LangChain** and **LangGraph**.
- Implemented **RAG** with **Azure AI Search**: document chunking, **Azure** embedding models, and a vector index for semantic retrieval.
- Built an autonomous SQL agent to generate, validate, and execute queries against **Azure MySQL Server** and **Azure Cosmos DB**.
- Deployed in a secure Azure environment: **Azure OpenAI**, **Azure Blob Storage**, **Azure VNet**, and private networking for secure data flow.
- Delivers real-time, accurate, grounded responses by combining LLMs, vector search, and database intelligence (production-ready architecture).

Network-Based Pattern Searching System

2024

- Developed a client-server pattern searching tool using **Python** and **JSON** for efficient text matching.
- Optimized error handling and file parsing, reducing query latency by 25% and improving reliability.

Self-Fuel Dispensing Automated Framework Using RFID Cards

Academic, 2023

- Designed and implemented an **RFID**-based automated fuel dispensing framework to minimize manual errors and unauthorized access.
- Improved transaction accuracy by 40% through real-time data synchronization and algorithm optimization.

EDUCATION

Bachelor of Technology in Electronics and Communication Engineering

Pragati Engineering College, Kakinada

Aug 2019 – Jun 2023

CGPA: 7.87 / 10

CERTIFICATIONS

- **Data Science Certification** – Cisco Systems (Generative AI Green Belt, 2025)
- **Cisco Certified Network Associate (CCNA)** (2024)
- **Cisco Certified DevNet Associate** (2023)
- **Python Programming** – NPTEL (2024)
- **Python (Basic)** – HackerRank (2024)
- **Automation Testing** – Test Automation Training (2024)

ADDITIONAL INFORMATION

Interests: AI/ML Applications, Test Automation, Full-Stack Development, Cloud Computing, and Generative AI